Information required for using the snapdrive snap restore command

Snapdrive for Unix

Ivana Devine, Aksel Davis
January 25, 2021

This PDF was generated from https://docs.netapp.com/us-en/snapdrive-unix/solaris/concept_informationrequired_forusing_the_snapdrive_snaprestore_command.html on August 06, 2021. Always check docs.netapp.com for the latest.

Table of Contents

Information required for using the snapdrive snap restore command 1

Information required for using the snapdrive snap restore command

To restore a Snapshot copy, determine which storage entity you want to restore, specify the name of the Snapshot copy, and so on.

The following table gives the information you need to supply when you use the `snapdrive snap restore` command.

Requirement/Option	Argument
Decide the type of storage entity that you wish to restore and enter that entity's name with the appropriate argument. • If you specify a host volume or file system to be restored, the argument you give is translated to the disk group containing it. SnapDrive for UNIX then restores the entire disk group. SnapDrive for UNIX freezes any file systems in host volumes in those disk groups and takes a Snapshot copy of all storage system volumes containing LUNs in those disk groups. • If you specify a file specification that is an NFS mountpoint, the argument translates to a directory tree. SnapDrive for UNIX restores only the NFS directory tree or file. Within the directory tree, SnapDrive for UNIX deletes any new NFS files or directories that you created after you created the Snapshot copy. This ensures that the state of the restored directory tree will be the same as when the Snapshot copy of the tree was made. • If you restore a LUN, SnapDrive for UNIX restores the LUN you specify. • If you restore a file system that is created directly on a LUN, SnapDrive for UNIX restores the LUN and the file system. • If the Snapshot copy contains storage entities that span multiple storage system volumes, you can restore any of the entities in that Snapshot copy.	
LUN (`-lun _file_spe_c`)	*name of the LUN. You must include the name of the storage system, volume, and LUN.*
Disk group (`-dg file_spec c`) or volume group (`-vg file_spec`)	*name of the disk or volume group*
File system (`-fs file_spec`)	*name of the file system*
File (`-file file_spec`)	*name of the NFS file*
Host volume (`-hostvol file_spec`) or logical volume (`-lvol file_spec`)	*name of the host or logical volume. You must supply both the requested volume and the disk group containing it; for example, - hostvol dg3/acct_volume.*

Requirement/Option	Argument
Specify the name for the Snapshot copy. If any of the *file_spec* arguments you supply at the command prompt currently exist on the local host, you can use a short form of the Snapshot copy name. If none of the *file_spec* arguments exist on the host, you must use a long form of the Snapshot copy name where you enter the storage system name, volume, and Snapshot copy name. If you use a long name for the Snapshot copy and the path name does not match the storage system and/or storage volume information at the command prompt, SnapDrive for UNIX fails. The following is an example of a long Snapshot copy name: `big_filer:/vol/account_vol:snap_20031115` Sometimes, the value supplied with the *file_spec* argument might not exist on the host. For example, if you had unmounted a file system or removed a disk group by exporting, deporting, or destroying, that file system or disk group could still be a value for the *file_spec* argument. It would, however, be considered a non-existent value. SnapDrive for UNIX can restore Snapshot copies for such a non-existent *file_spec*, but you must use the long Snapshot copy name. When you restore Snapshot copies that span multiple storage systems and volumes, and contain a nonexistent *file_spec* argument, SnapDrive for UNIX permits an inconsistency in the command line. It does not allow for existing *file_spec* arguments. If you want to restore only one storage entity from a multiple storage system Snapshot copy, the Snapshot copy you specify does not need to be on the same storage system as the storage system containing the storage entity. The short form of the same Snapshot copy name would omit the storage system and storage system volume name, so it would appear as: *snap_20031115*	
Snapshot copy name (`-snapname`)	*snap_name*
It can be either a short name, such as *mysnap1*, or a long name that includes the storage system name, volume, and Snapshot copy name. Generally, NetApp recommends that you use the short name. If the *file_spec* argument is non-existent: that is, it no longer exists on the host; see the explanation of the *file_spec* argument. Then you must use the long name for the Snapshot copy.	
`-reserve | -noreserve`	
Optional: If you want SnapDrive for UNIX to create a space reservation when you restore the Snapshot copy.	
`-force`	~
`-noprompt`	~

Requirement/Option	Argument
Optional: Decide if you want to overwrite an existing Snapshot copy. Without this option, this operation halts if you supply the name of an existing Snapshot copy. When you supply this option and specify the name of an existing Snapshot copy, it prompts you to confirm that you want to overwrite the Snapshot copy. To prevent SnapDrive for UNIX from displaying the prompt, include the `-noprompt` option also. (You must always include the `-force` option if you want to use the `-noprompt` option.) You must include the -force option at the command prompt if you attempt to restore a disk group where the configuration has changed since the last Snapshot copy. For example, if you changed the way data is striped on the disks since you took a Snapshot copy, you would need to include the -force option. Without the `-force` option, this operation fails. This option asks you to confirm that you want to continue the operation unless you include the `-noprompt` option with it. If you added or deleted a LUN, the restore operation fails, even if you include the `-force` option.	
`-devicetype`	~
Optional: Specify the type of device to be used for SnapDrive for UNIX operations. This can be either “shared” that specifies the scope of LUN, disk group, and file system as host cluster-wide or “dedicated” that specifies the scope of LUN, disk group, and file system as local. If you specify the `-devicetype` dedicated option, all the options of `snapdrive restore connect` command currently supported in SnapDrive 2.1 for UNIX function as they always have. If you initiate the `snapdrive restore connect` command with the `-devicetype shared` option from any nonmaster node in the host cluster, the command is shipped to the master node and executed. For this to happen, you must ensure that the `rsh` or `ssh` access-without-password-prompt for the root user should be configured for all nodes in the host cluster.	
mntopts	~
Optional: If you are creating a file system, you can specify the following options: • Use `-mntopts` to specify options that you want to pass to the host mount command (for example, to specify host system logging behavior). The options you specify are stored in the host file system table file. Allowed options depend on the host file system type. • The *`-mntopts`* argument is a file system `-type` option that is specified using the mount command `-o` flag. Do not include the `-o` flag in the *`-mntopts`* argument. For example, the sequence -mntopts tmplog passes the string `-o tmplog` to the `mount` command, and inserts the text tmplog on a new command line. If you pass any invalid *`-mntopts`* options for storage and snap operations, SnapDrive for UNIX does not validate those invalid mount options.	

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.